exist nonzero codes $\mathbf{h}$ for which $\mathbf{Wh} = 0$ (i.e. $\ker(\mathbf{W}) \neq \{\mathbf{0}\}$ since $\dim\ker(\mathbf{W}) = d_h - \operatorname{rank}(\mathbf{W}) \geq d_h - d_y > 0$). Thus, information is inevitably lost, so multiple codes $\mathbf{h}$ can produce the same activation $\mathbf{y}$, resulting in its infinitely many possible reconstructions.

This non-uniqueness has a deeper consequence in that there is nothing to identify since identifying a ground truth solution would imply presupposing its uniqueness. But, the model $\mathbf{y} = \mathbf{Wz}$ is inherently not identifiably since, for any invertible (d_z by d_z) matrix $\mathbf{M}$, we can write $\mathbf{y} = (\mathbf{WM})(\mathbf{M}^{-1}\mathbf{z}) = \mathbf{W}'\mathbf{z}'$, defining a different dictionary and latent variable that result in the same observed variable $\mathbf{y}$. Thus, interpreting $\mathbf{h} = r(\mathbf{y})$ as recovering latents requires an additional selection principle that prefers one solution among many consistent codes.

To obtain a unique solution, one typically restricts the code to be sparse, mirroring the latent sparsity assumption in Equation (1). Concretely, we restrict $\mathbf{h} \in \mathbb{R}^{d_h}$ to lie in the k-sparse set [1]:

$$\Sigma_k^{(d_h)} \;:=\; \{\mathbf{h} \in \mathbb{R}^{d_h} : \|\mathbf{h}\|_0 \leq k\},$$

This sparsity can be enforced softly (e.g., via an ℓ_1 penalty) or as a hard constraint through constrained optimisation (Ramirez et al., 2025; Gallego-Posada et al., 2025), or as implemented in TopK SAEs, which retain only the k largest-magnitude coordinates of $\mathbf{h}$ per input. Algebraically, sparsity replaces the unconstrained feasibility set $\{\mathbf{h} \in \mathbb{R}^{d_h} : \mathbf{Wh} = \mathbf{y}\}$ (typically infinite) with the constrained set $\{\mathbf{h} \in \Sigma_k^{(d_h)} : \mathbf{Wh} = \mathbf{y}\}$, which can be a singleton under suitable conditions on $\mathbf{W}$. In this sense, $r(\mathbf{y})$ is meaningful only insofar as it implements a consistent sparse selection of dictionary atoms among the many codes that reconstruct the same $\mathbf{y}$.

Identifiability of sparse codes h. Restricting the codes to be sparse is not sufficient to guarantee uniqueness—there may still exist distinct $\mathbf{h}, \mathbf{h}' \in \Sigma_k^{(d_h)}$ with $\mathbf{Wh} = \mathbf{Wh}'$. We need a property of the dictionary $\mathbf{W}$ ensuring that it does not collapse sparse codes onto each other so that they can be identifiable. A standard sufficient condition is the restricted isometry property (RIP) (Candes and Tao, 2006; Donoho, 2006b; Candès and Wakin, 2008): $\mathbf{W}$ satisfies RIP of order s with constant δ_s if $(1-\delta_s)\|\mathbf{h}\|_2^2 \leq \|\mathbf{Wh}\|_2^2 \leq (1+\delta_s)\|\mathbf{h}\|_2^2$ $\forall \|\mathbf{h}\|_0 \leq s$. Intuitively, $\mathbf{W}$ approximately preserves the geometry of the sparse codes $\mathbf{h}$, neither inflating nor collapsing them beyond a tolerance determined by the number of non-zero components, i.e., sparsity s of $\mathbf{h}$. When we consider $\mathbf{h} \in \Sigma_k^{d_h}$, we want to ensure that two distinct k-sparse codes $\mathbf{h}$ and $\mathbf{h}'$ are distinguishable, i.e., their difference is not invisible to $\mathbf{W}$. Since, their difference can have at most $2k$ non-zeros, this is equivalent to asking that $\mathbf{W}$ maps no nonzero $2k$-sparse vector to zero, since if $\mathbf{Wh} = \mathbf{Wh}'$, then $\mathbf{W}(\mathbf{h}-\mathbf{h}') = 0$. RIP at order $2k$ ensures exactly this. [2] So, under RIP, it is impossible for two different sparse vectors to map to the same output since their difference cannot be in the null space of $\mathbf{W}$, ensuring injectivity. RIP is fulfilled for random Gaussian matrices projecting down into $d_y \geq O(k \ln(\frac{d_h}{k}))$ dimensions (Candès and Wakin, 2008).

Implication: How many concepts can we encode?

The RIP bound $d_y \geq O(k \ln(d_h/k))$ is *logarithmic* in the dictionary size d_h, meaning the number of latent concepts can be exponentially larger than the activation dimension. A typical value of a constant to realise the bound is 2, i.e. $d_y \geq 2(k \ln(d_h/k))$ (Baraniuk et al., 2008). With $d_y = 4{,}096$ and $k = 50$ active concepts, the bound permits dictionaries as large as $d_h \approx e^{d_y/2k} \sim e^{40}$—astronomically more concepts than dimensions. Even conservatively, a transformer hidden size of 4,096 can in principle support unique recovery over millions of latent concepts, provided only a few dozen are active at once. Thus, in the case of dictionaries with up to 16 million features on GPT-4 activations (Gao et al., 2024), or 34 million features with Claude 3 Sonnet (Templeton, 2024), $d_h = 10^6 - 10^7$ is still in the feasible range. The practical takeaway is that scaling up the dictionary is essentially free from the perspective of compressed sensing theory.

It is important to distinguish two levels of identifiability that arise in this setting. The first is *code-level identifiability*, i.e., given a fixed and known dictionary $\mathbf{W}$, when can we uniquely recover the sparse code $\mathbf{h}$ from an observation $\mathbf{y}$. This is precisely what RIP guarantees—it ensures that per-sample inference methods such as basis pursuit (Chen et al., 2001), or ISTA (Daubechies et al., 2004)/FISTA (Beck and Teboulle, 2009) converge to the unique k-sparse solution consistent with $\mathbf{y}$. The second is *dictionary identifiability*: can we recover the dictionary $\mathbf{W}$ itself from observations $\mathbf{y}$? Classical results show that $\mathbf{W}$ is identifiable when the data is sufficiently sparse and diverse (Hillar and Sommer, 2015; Gribonval et al., 2015), which is similar to a more flexible sufficient support variability condition on data for learning the dictionary (Joshi et al., 2025). These guarantees, however, are only as useful as the optimisation procedure that realises them.

Sparse Coding and Amortisation. Through point-wise inference, sparse coding infers a sparse code per input. Concretely, given samples $\{\mathbf{y}_i\}_{i=1}^p$ and fixed dictionary $\mathbf{W}$, the canonical *pointwise* formulation solves, for each i,

$$\mathbf{h}_i^\star \in \arg\min_{\mathbf{h}\in\mathbb{R}^{d_h}} \tfrac{1}{2}\|\mathbf{y}_i - \mathbf{Wh}\|_2^2 + \lambda\|\mathbf{h}\|_1,$$

(or equivalently, a hard sparsity constraint $\mathbf{h} \in \Sigma_k^{(d_h)}$).

[1] equivalently requiring that $\operatorname{supp}(\mathbf{h}) \in \mathcal{S}_k^{(d_h)} := \{S \subseteq [d_h] : |S| \leq k\}$.

[2] A classical alternative is $\operatorname{spark}(\mathbf{W}) > 2k$ (spark of a matrix is the smallest number of its columns that are linearly dependent), no linear combination of $2k$ or fewer columns can weighted by $\mathbf{W}$ can sum to zero, i.e., $\ker(\mathbf{W}) \cap \Sigma_{2k}^{(d_h)} = \{\mathbf{0}\}$ and hence $\mathbf{W}$ is injective on $\Sigma_k^{(d_h)}$ (Donoho and Elad, 2003; Gribonval and Nielsen, 2004). But spark is typically intractable to compute.

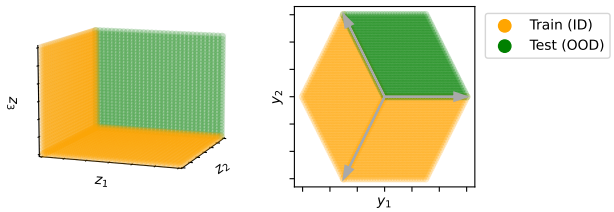

Figure 2: Compositional OOD split. *Left:* In-distribution (ID) training data covers support pairs (z_1, z_2) and (z_2, z_3) and the novel combination (z_1, z_3) is held out for OOD evaluation. *Right:* Same split in activation space $\mathbf{y} = \mathbf{W}\mathbf{z}$.

Algorithms such as Iterative Shrinkage-Thresholding Algorithm (ISTA) and FISTA compute $\mathbf{h}_i^\star$ by iterating a sequence of code estimates $\mathbf{h}_i^{(0)}, \mathbf{h}_i^{(1)}, \ldots$ for each input, typically starting from $\mathbf{h}_i^{(0)} = \mathbf{0}$ and applying repeated gradient-and-thresholding updates until convergence (Beck and Teboulle, 2009; Daubechies et al., 2004).

When the dictionary is unknown, inference is paired with *dictionary learning* (Olshausen and Field, 1996; Mairal et al., 2010), to alternate between (i) estimating $\{\mathbf{h}_i\}_{i=1}^p$ given the current $\mathbf{W}$ and (ii) updating $\mathbf{W}$ to minimise reconstruction error under a column-norm constraint. A standard formulation is,

$$\min_{\mathbf{W}, \{\mathbf{h}_i\}_{i=1}^p} \sum_{i=1}^{p} \left(\tfrac{1}{2} \|\mathbf{y}_i - \mathbf{W}\mathbf{h}_i\|_2^2 + \lambda \|\mathbf{h}_i\|_1 \right) \quad \text{s.t.} \|\mathbf{W}_{:,j}\|_2 \leq 1 \quad \forall j \in [d_h]. \tag{3}$$

Implication: How sparse must the codes be?

k enters the RIP bound roughly linearly, denoting the number of interpretable concepts simultaneously encoded in the activation vector. It is unclear a priori what the expected value of k would be across different activations. Rearranging the bound gives the maximum feasible sparsity: $k_{\max} \approx d_y / \ln(d_h/k)$. For $d_y = 4{,}096$ and $d_h = 10^7$, this yields $k_{\max} \approx 185$, depending on the constant. Beyond this, no inference procedure—per-sample or amortised—can guarantee unique recovery. The choice of k in training SAEs therefore has implications beyond the reconstruction–sparsity trade-off as it determines whether the problem is even theoretically solvable.

In contrast, amortised methods replace per-input iterative solving with a feed-forward encoder that predicts $\mathbf{h}$ in a single forward pass, learning to approximate the solution across a training distribution(Vafaii et al., 2024, 2025). We can have amortised sparse inference (e.g., LISTA (Gregor and LeCun, 2010)) that unrolls ISTA iterations into learnable layers, so that its architecture is structurally tied to the sparse-coding objective and the dictionary is treated as given. SAEs, the dominant amortised approach in interpretability, are amortised autoencoding: they jointly learn both the encoder and the dictionary (the decoder q). An SAE encoder is free to learn any mapping that minimises reconstruction loss on the training data, including solutions that exploit distributional shortcuts rather than performing principled sparse decomposition.

Implication: How to obtain sparse codes?

In the fully unsupervised setting assumed while training SAEs, there is no guarantee that the underlying latent codes are sparse enough for recovery. This connects to a broader impossibility: Hyvärinen and Pajunen (1999); Locatello et al. (2019) show that fully unsupervised disentanglement is impossible without inductive biases on the model or the data. In the compressed-sensing framing, the required inductive bias is precisely sufficient sparsity: the data must be generated by activations involving few enough concepts at once. Joshi et al. (2025) propose to leverage concept shifts being sparser to effectively automate the creation of sufficiently sparse and diverse observations from model activations.

In the fully unsupervised setting, the general impossibility of unsupervised identifiabilty implies typical SAEs have been shown not to be identifiable (O'Neill et al., 2024). We hypothesise that this is one reason for the poor generalisation of SAEs OOD found in the literature (Kantamneni et al., 2025). In principle, a disentangled generalisation should allow for better OOD generalisation on downstream tasks (Schölkopf* et al., 2021)—but whether this promise holds in practice depends on the entire pipeline—from dictionary quality, through the inference procedure, to the downstream task. In this paper, we focus on this downstream question. We operationalise the central question from Section 1 by studying compositional distribution shifts (novel combinations of known concepts) which decompose it into two testable research questions:

i **Is sparse inference even necessary?** Linear probes outperforming SAEs on OOD tasks does not mean linear decoding is sufficient, it just means amortised inference is failing. Labels that are linearly separable in latent space become nonlinearly separable in activation space, and this nonlinearity is exposed precisely under compositional shifts. Rather than abandoning sparse coding for linear probes, we suggest the solution lies within the compressed-sensing framework—but the bottleneck may not be where one expects.

ii **What is the bottleneck: inference or dictionary learning?** SAEs jointly learn a dictionary and an encoder. We decompose their failure to ask whether per-sample inference with an SAE-learned dictionary can close the gap, and if not, whether the dictionary itself—rather than the encoder—is the binding constraint.

Compositional generalisation. We evaluate inferred sparse

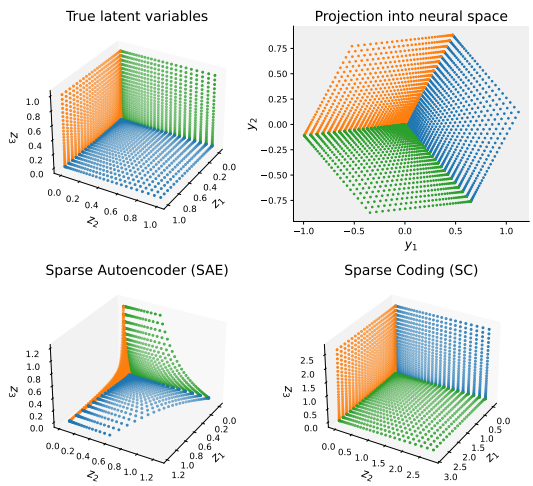

Figure 3: SAEs fail to recover latent variables under superposition, but sparse coding succeeds. **Top left:** Ground-truth latents ($d_z = 3$, $k = 2$); colors denote active-variable combinations. **Top right:** Activation space $\mathbf{y} = \mathbf{W}\mathbf{z}$ ($d_y = 2$); factors overlap after projection. **Bottom left:** SAE reconstruction; planes are not recovered. **Bottom right:** Sparse coding reconstruction; latents are identified up to scaling.

codes via a downstream binary prediction task, where the target t depends on the latent $\mathbf{z}$ through $t = u(\mathbf{z}) \in \{0, 1\}$; hence, predicting t from $\mathbf{h}$ through $\hat{t} = v(\mathbf{h})$ is a standard supervised proxy for testing whether $\mathbf{h}$ preserves task-relevant information about $\mathbf{z}$, such as by fitting a logistic head on $\mathbf{h}$ and evaluating it through the log-odds $\log \frac{\Pr(t=1|\mathbf{h})}{\Pr(t=0|\mathbf{h})} = \mathbf{a}^\top \mathbf{h} + a_0$ for weights $\mathbf{a}$.

Train and test sets differ in which combinations of generative factors co-occur (Figure 2). In terms of supports $S \subseteq [n]$ of the sparse latent $\mathbf{z}$ (cf. Equation (1)), the ID data excludes a structured subset of support patterns while the OOD data concentrates on those withheld combinations. E.g., consider $k = 2$, and three factors $z_1.z_2, z_3$. The ID data contains support pairs (z_1, z_2) and $(z_2, z_3$ while the combination (z_1, z_3) is held out for OOD evaluation (Figure 2. A donwstream label $t = \mathbf{1}\{z_1 > 0.5\}$ depends only on z_1, but a model trained ID may learn a shortcut: since z_1 always co-occurs with z_2, predictor $\mathbf{a}^T\mathbf{h}+a_0$ can achieve high accuracy by tracking features correlated with z_2 instead of isolating z_1 itself. The OOD split breaks this shortcut by pairing z_1 with z_3 instead (Figure 1). This follows the withheld-co-occurrence logic central to spurious-correlation benchmarks like Waterbirds (Sagawa et al., 2019). Achieving OOD success therefore requires the encoder to recover coordinates where the evidence for t remains stable across different latent factor recombinations. Figure 3 illustrates a toy example example of this failure, where SAEs fail to reconstruct the OOD plane (z_1, z_3), while sparse coding reconstructs all latents.

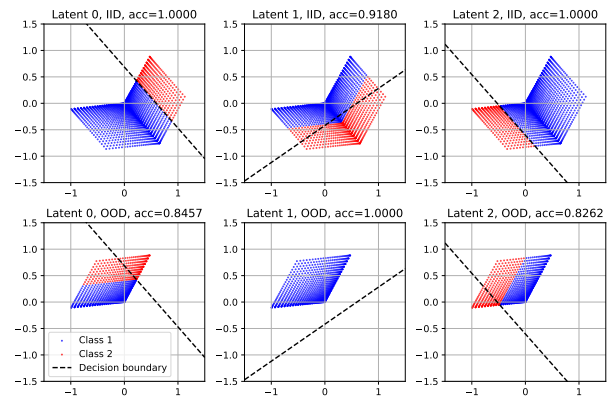

Figure 4: **Linear probes fail OOD under overcompleteness.** Each column sets $t = z_i$. The linear classifier fits the ID decision boundary well, but the compression $\mathbf{y} = \mathbf{A}\mathbf{z}$ introduces nonlinearity that is only exposed OOD, causing catastrophic generalisation failure (columns 1, 3) or, even, poor ID accuracy (column 2).

Linear Probing under the LRH. When $\mathbf{W}$ is injective ($d_z \leq d_y$), linear separability is invariant: $\mathbf{W}$ can rotate or rescale latent space, but a label that is linear in $\mathbf{z}$ remains linear in $\mathbf{y} = \mathbf{W}\mathbf{z}$ (Figure 1a; also (Garg et al., 2026) Fig. 1).

However, under overcompleteness, even labels that are linearly separable in latent space can become nonlinearly separable in activation space. When $\mathbf{W}$ is non-injective, multiple $\mathbf{z}$ map to the same $\mathbf{y}$, potentially lying on opposite sides of the latent hyperplane; in that case, no *linear* rule in $\mathbf{y}$ can match the latent separator without error. This is the geometric phenomenon illustrated in Figure 1b: a hyperplane separator in the full latent space can appear linear on the in-distribution slice of observed mixtures, yet become effectively nonlinear (or even ill-posed [3]) in activation space after being transformed through $\mathbf{W}$. Figure 4 illustrates different failure cases of linear probes under this compositional setting. Under sparse coding, RIP does not make $\mathbf{W}$ globally invertible, but it ensures that $\mathbf{W}$ does not collapse *sparse* directions, so that sparse latents remain (nonlinearly) distinguishable and pointwise sparse inference is well-posed.

4 EXPERIMENTS

We investigate two questions: (i) is the OOD failure of SAEs a fundamental limitation of sparse inference under superposition, or is it a failure of specific components (dictionary, encoder, or both)? (ii) If the problem is solvable in principle, what is the bottleneck in practice? We use an oracle baseline—FISTA with the ground-truth dictionary—to establish an upper bound on what per-sample inference can achieve, then progressively decompose the gap between this oracle and SAEs. Methods span four families from fully

[3] Ill-posedness arises when $\exists\, \mathbf{z} \neq \mathbf{z}'$ with $\mathbf{W}\mathbf{z} = \mathbf{W}\mathbf{z}'$ but $t(\mathbf{z}) \neq t(\mathbf{z}')$, in which case t is not a function of $\mathbf{y}$ at all.